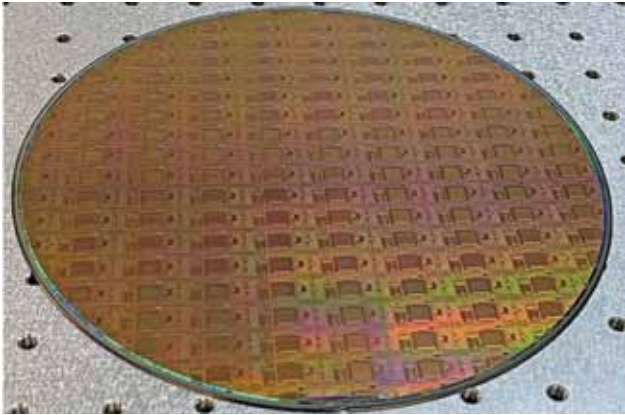


the label of the closest training data point, and the third uses a weighted average. Only the third method results in optimal performance, and a set of activation functions were found to always use it. They tested this theory on several classification benchmarking tasks and found that it led to improved performance in many cases.

Merger of silicon with memristor enables energy-efficient memory chip

A team of researchers from the University of Southern California and the University of Massachusetts has developed a protocol for devices to reduce ‘noise’ and demonstrated the

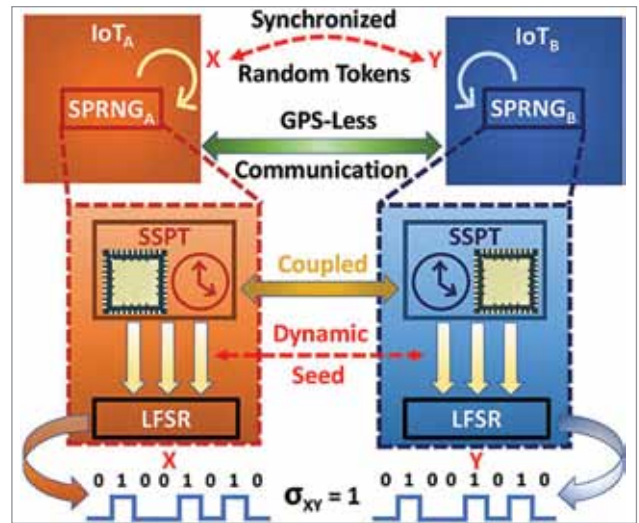


New chip design to provide greatest precision in memory to date, will enable powerful AI in your portable devices (Credit: Joshua Yang of USC and TetraMem)

practicality of using this protocol in integrated chips. The memory chip has a record-breaking information density of 11 bits, surpassing all previous memory technologies. These small but potent chips could significantly boost the power of mobile devices, serving as both memory and processor. Millions of these chips could work together to perform AI tasks and require only a small battery. New device merges silicon with metal oxide memristors to make potent, energy-efficient chips. Atom positions (rather than electrons) represent data, allowing more information storage in analogue form. Processing occurs where the data is stored, removing the ‘von Neumann bottleneck’ present in traditional systems.

Prototype developed to better secure communications using SPRNG

Researchers from Washington University in St. Louis have developed a prototype method to better secure communications using a synchronised pseudo-random-number generator (SPRNG). Researchers created a prototype synchronised



The concept of GPS-free secure communication in spatially separated IoTs with SPRNGs: The IoTs generate random tokens using the SPRNG for use as cryptographic keys. The tokens are generated using a combination of a fast, low-complexity LFSR seeded by the secure self-powered timers (SSPT) (Credit: Frontiers in Computer Science, 2023)

with self-powered timer array using quantum-mechanical tunneling of electrons that is secure against tampering, snooping, and side-channel attacks. They utilised Fowler-Nordheim quantum tunneling, which simplifies the process of creating an energy-efficient and secure connection for computer modeling. It modifies the triangular barrier shape as electrons jump through it, making it more self-powered and resistant to attacks than other methods. The SPRNG method can act as a trusted platform module for IoT, authenticating secure transactions, as per researchers.

Plant based film that gets cooler in sunlight comes in different textures and colours

A team of researchers have developed an eco-friendly alternative—a plant based film that gets cooler when exposed to sunlight and comes in a variety of textures and bright, iridescent colours. Passive daytime radiative cooling (PDRC) allows its surface to emit heat into space, making it colder than surrounding air. It reduces the need for power-hungry cooling methods in buildings. While most PDRC paints and films under development are white or mirrored, structural colour can provide colour without pigments. Soap bubbles demonstrate structural colour, displaying a prism of hues on their surfaces due to the varying thickness of their film and the way light interacts with it. Vignolini studied natural structural colours and discovered that CNCs, derived from plant cellulose, can create colourful films without added pigment and promote PDRC due to cellulose’s natural properties.